

Dimension-Free Diffusion for Approximate Gaussian Mixtures

Independent CPU numerical audit · three finite checks · not a proof replacement

Paper: Gen Li, Changxiao Cai, Yuting Wei · Audit revision: arxiv:2504.05300v1



Evidence & rerun
Open the Trackio logbook

Within deliberately narrow synthetic settings, all three checks support their target behavior. Controls separate when assumptions are relaxed.

0

analytic variance-error range across d=1...64

linear

mean shift under additive score error

3.6e−15

range of max tr(I+J) across dimensions

01 · PAPER CONTEXT

What the theorem says

For targets close in total variation to a unit-covariance isotropic GMM, the paper bounds DDPM error by discretization, score-estimation, and approximation terms.

$$\text{TV}(X_0, Y_1) \lesssim \log^2(KT) \log^2(T)/T + \varepsilon_{\text{score}} \sqrt{\log(T)} + \sqrt{(d \varepsilon_{\text{approx}}) \log^3(T) (dT)}$$

- Every behavior test visibly failed before its code was added.
- Analytic identities are checked independently by finite differences where applicable.
- JSON and CSV evidence is byte-identical in repeated executions.

Context only: a paraphrase of Theorem 1 from the pinned paper source; paper values are never reproduced measurements.

Conditions kept visible

- GMM components have covariance I_d .
- Component means obey the paper’s polynomial norm condition.
- Score error is an average L^2 error over time.
- The theorem uses TV distance; these checks use finite diagnostic proxies.

Evidence boundary

This synthetic CPU audit trains no score network and estimates no end-to-end TV distance. It does not establish the paper’s asymptotic complexity.

02 · COMPUTED OUTPUT

One-step discretization

At the stationary standard Gaussian with exact score $-x$, one Euler/DDPM-like step has coordinate variance $1+h^2$. With $h=0.125$, the analytic per-coordinate error is 0.015625 for every tested dimension.

dimension	analytic error	empirical diagnostic
1	0.015625	0.06203
4	0.015625	0.04681
16	0.015625	0.01685
64	0.015625	0.01130

Independent identity

$$x' = (1-h)x + \sqrt{(2h)}z \Rightarrow \text{Var}(x') = 1+h^2$$

The identity holds coordinate-wise. The empirical column is retained to expose finite-sample variation, while the claim decision uses the analytic range.

Scope verdict: toy support

The zero analytic range audits dimension independence in a single exact-score Gaussian step. It does not reproduce the theorem’s complete multi-step TV guarantee.

- Threshold: analytic range $\leq 1e-12$.
- Observed range: exactly 0.
- Dimensions: 1, 4, 16, 64.

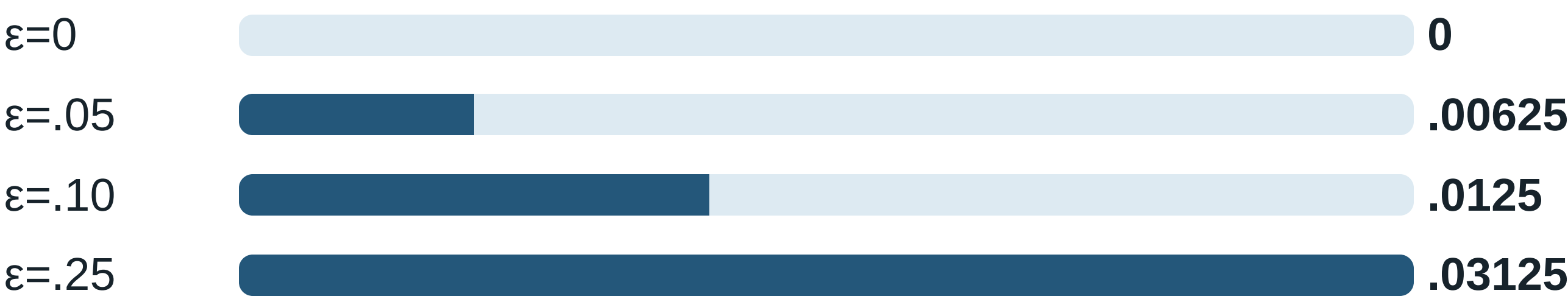
The empirical deviations shrink irregularly with dimension because they are Monte Carlo diagnostics, not the analytic decision statistic.

Seed 2026 · 2,048 samples per dimension · CPU execution only

03 · COMPUTED OUTPUT

Controlled score error

A constant score perturbation with L^2 norm ε shifts the one-step mean by $h\varepsilon$. The measured analytic shifts are monotone and exactly separable in this construction.



Decision rule

The analytic mean shift must be nondecreasing with perturbation norm. Here it follows $h\varepsilon$ exactly at $h=0.125$, isolating the score-error term from sampling noise.

The empirical mean norm is logged as a diagnostic but is not substituted for the analytic controlled construction.

The paper’s theorem permits time-averaged L^2 score error; this one-step construction audits only the separability and monotonicity of a calibrated additive perturbation.

Misspecification control

Replacing the calibrated constant perturbation with a state-dependent cubic term violates the controlled-error setup. Its empirical second moment rises to 1.393.

- Control: $0.5x^3$ state-dependent perturbation.
- Observed empirical mean norm: 0.218.
- Recorded separately with `assumption_satisfied=false`.

04 · COMPUTED OUTPUT

The trace that matters

For a symmetric two-component unit-covariance GMM embedded in dimensions 1, 4, 16, and 64, the audit computes the paper-aligned quantity $\text{tr}(I+J)$.

d	max tr(I+J)	FD error
1	3.9999389	5.6e−10
4	3.9999389	6.1e−10
16	3.9999389	6.9e−10
64	3.9999389	9.2e−10

Why dimension cancels

The component means occupy one fixed active coordinate. For unit covariance, the ambient $-I$ term cancels inside $\text{tr}(I+J)$, leaving the posterior covariance of component means.

$$\text{tr}(I+J) = \text{tr}(\text{Cov}[\mu_k \mid x])$$

The implementation compares the analytic trace against centered finite differences at 2,048 deterministic evaluation points.

Unit-covariance control

Setting component variance to 0.05 breaks the unit-covariance setup and makes $\text{tr}(I+J)$ strongly dimension-dependent. The baseline max range is 3.6e−15.

- Baseline variance: 1.0.
- Control variance: 0.05.
- Control mean $\text{tr}(I+J)$ changes from -9.0 at $d=1$ to -1206.0 at $d=64$.

The analytic trace is checked independently by centered finite differences.